

Apple of My Earth (11')

A Generative Film About Protection Gaps
(and Potatoes)

CAS AICP Final Project
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What is the Protection Gap?

The **gap** between:

- Risk exposure
- Actual protection

...across **systems**:

- Natural
- Human
- Technological

-> **Systems** reduce risk, but never eliminate it.



The Narrative

Three **Acts** - one **entity**:

- Colca Valley (Peru, ~1450, late Inca period)
- Einsiedeln (Switzerland, 1936, before flooding of Sihl Valley)
- Mars (~2100 - first permanent colony)

Main **protagonist**:

The potato plant ...

...in all its life stages, constant and at the center of every frame

Theme:

The protection gap, its nuances and dramatic qualities: environmental influences and human intervention



Visual Language

Envisioned **cinematic rules**:

- Camera locked to plant (center frame). The world changes around it
- Foreground in focus / background blurred
- Time unfolds in one fixed perspective at different speeds
- System-defining events unfold in the background (e.g. a volcanic eruption)
- Humans only partially visible or out of focus
- Natural life cycle visible (growth, harvest, regrowth)

Style:

- Photorealistic
- cinematic, documentary-like
- aiming to evoke the feeling of time-travel
- Sound effects only, no music

Technical, aesthetic and ethical shortcomings of the generated outcome, e.g. due to model-use, and/or budget constraints will be embraced as flaws of current GenAI workflows.

The project thesis paper will then analyze these shortcomings and discuss them critically.

The screenplay (completed)

11 pages, author-driven, self-written. No GenAI use for creative ownership

Unconstrained imagination, in disregard of limitations of AI video generation

Serves as project's primary source data ("narrative data")

LLM / project agents used for research and translation tasks, while conceptualizing the production pipeline.

SCRIPT TITLE

Written by

Name of First Writer

Based on, If Any

Address
Phone Number

The screenplay (sample)

Explorative Storyboarding (completed)

- **Project information** accumulated in LLM / project agent turned into reference still imagery.
- **Goal:** Gather learnings to inform generative process for video pipeline
- Identification of **story beats**
- **Learnings** about required layering and object control
- **Learnings** of pitfalls in GenAI image generation (temporal progress vs consistency)



Storyboarding using LLMs (ChatGPT 5.5, “thinking”)

Positive outcomes

- Fast and relatively easy visualization based on writing and production conversations (agents)
- Filmic direction works, e.g. “add motion blur”

Challenges:

- Directionality of objects (e.g. bee flies off to the left) remains a challenge
- Unintended abstractification of elements (soil turns to plastic texture, “AI slop”)
- Pixel precision is impossible, leading to stop motion aesthetic
- Layering and masking will be a necessity for precise direction (layers get overridden with temporal progression)

The “beat” JSON (completed) (1/2)

Every **shot** in the film is one **structured record**.

A **beat** is a small block of data made up of fields - each one a question the system needs answered before it can render the shot.

The fields fall into five groups:

1. Identity

- beat_id — which beat in which act
- duration — how long it lasts on screen

2. The plant (the protagonist)

- stage — sprout, mature, flowering, harvested, charred, submerged
- condition — healthy, trembling, damaged, regrowing
- position — locked anchor in the frame

```
{
  "beat_id": "A1.B003",
  "beat_name": "Flowering and bee visit",
  "beat_order": 3,
  "act_id": "A1",
  "script_excerpt": "As the sun embraces it with warm light, the plant shows gratitude by developing a beautiful, violet flower... The bloom attracts a curious BEE, hovering into frame... drinking nectar from the flowers!.",
  "script_summary": "Plant develops violet flowers under warm afternoon light. A bee enters, inspects, and feeds from the blooms. As the sun sets, the bee leaves, the flowers fade, leaves thicken, soil swells.",
  "timing": {
    "duration_sec": 8,
    "time_of_day": "accelerated",
    "tempo": "medium",
    "transition_in": "cut",
    "transition_out": "cut"
  },
  "composition_strategy": "layered_composite",
  "background_plate": {
    "plate_id": "A1_plate_pristine",
    "modifications": ["warm_afternoon_grade"]
  },
  "foreground_anchor": [0.5, 0.78],
  "layers": [
    {
      "layer_id": "plant_flowering",
      "type": "foreground_subject",
      "depth_band": "foreground",
      "focus_state": "sharp",
      "z_index": 10,
      "source": "render_motion",
      "prompt": "mature potato plant developing violet flowers with yellow centers, gentle warm late-afternoon light, sharp focus, locked center",
      "alpha_strategy": "auto_segment",
      "blend_mode": "normal",
      "opacity": 1.0,
      "motion": "flower_bloom_then_fade"
    },
    {
      "layer_id": "bee",
      "type": "foreground_actor",
      "depth_band": "foreground",
      "focus_state": "sharp",
      "z_index": 11,
      "source": "render_motion",
      "prompt": "single honeybee hovering near violet potato flower, motion-blurred wings, sharp body, feeding pose",
      "alpha_strategy": "auto_segment",
      "blend_mode": "normal",
      "opacity": 1.0,
      "motion": "hover_feed_then_leave"
    },
    {
      "layer_id": "make_plume_calm",
      "type": "event_effect",
      "depth_band": "background",
      "anchor_in_plate": true,
      "anchor_point_id": "jabsckaya_humil",
      "blend_mode": "screen",
      "opacity": 0.7
    }
  ],
  "plant_state": {
    "present": true,
    "stage": "flowering_to_post_flower",
    "condition": "healthy",
    "flowers_visible": true
  },
  "animal_presence": {
    "present": true,
    "species": ["bee"],
    "visibility_mode": "full_but_small",
    "action": "pollination"
  },
  "environment_state": {
    "location_variant": "endian_valley_pristine",
    "light_state": "warm_afternoon_to_dusk"
  },
  "event_state": {
    "event_active": false,
    "system_status": "stable"
  },
  "sound_state": {

```

The “beat” JSON (completed) (2/2)

3. The **world** (what stays still)

- background_plate — which locked background this beat uses
- plate_variant — pristine, terraced, destroyed, decaying (same topography, different state)
- depth_bands — what lives in foreground, midground, background

4. The **event** (what changes)

- layers — smoke, water, dust, light, each with its own anchor and behavior

5. The **grammar** (what the system enforces)

- camera — locked, no movement
- aspect_ratio — 16:9
- composition_rules — plant centered, humans partial, foreground sharp

```
{
  "plant_state": {
    "present": true,
    "stage": "flowering_to_post_flower",
    "condition": "healthy",
    "flowers_visible": true
  },
  "animal_presence": {
    "present": true,
    "species": "llama",
    "visibility_mode": "full_but_small",
    "action": "pollination"
  },
  "environment_state": {
    "location_variant": "andesian_valley_pristine",
    "light_state": "warm_afternoon_to_dusk"
  },
  "event_state": {
    "event_active": false, "system_status": "stable"
  },
  "sound_state": {
    "soundscape_id": "andesian_valley_drylight_soundscape",
    "foreground_sounds": ["Two_Boys"]
  },
  "continuity_contract": {
    "must_match_plate_pixel_perfect": "A1_plate_pristine",
    "continuity_reference_beat": "A1_0000",
    "next_beat_id": "A1_0004"
  }
},
{
  "beat_id": "A1_0004",
  "beat_name": "Night falls, llama approaches",
  "beat_order": 4,
  "act_id": "A1",
  "script_excerpt": "Night falls and a bright moon rises, shedding cold light... we begin to hear SOUNDS of toe-nailed paws approaching. And after a moment, four sure-footed, slender legs enter the frame, slowly approaching the potato plant... A LLAMA!",
  "script_summary": "Night, moonlit valley. Approaching toe-nailed paw sounds. Llama enters partially - slender legs, lowered head sniffing the plant.",
  "timing": {
    "duration_sec": 7,
    "time_mode": "real_time",
    "tempo": "slow",
    "transition_in": "fade_to_night",
    "transition_out": "cut"
  },
  "composition_strategy": "layered_composite",
  "background_plate": {
    "plate_id": "A1_plate_pristine",
    "modifications": ["night_moonlight_grade", "cool_silver_blue_tint"]
  },
  "foreground_anchor": [0.5, 0.78],
  "layers": [
    {
      "layer_id": "plant_night",
      "type": "foreground_subject",
      "depth_band": "foreground",
      "focus_state": "sharp",
      "z_index": 10,
      "source": "render_motion",
      "prompt": "neon potato plant at night, edge-lit by cool moonlight, post-flower swollen leaves, sharp focus, locked center",
      "alpha_strategy": "auto_segment",
      "blend_mode": "normal",
      "opacity": 1.0
    },
    {
      "layer_id": "llama_partial",
      "type": "foreground_actor",
      "depth_band": "midground",
      "focus_state": "softening",
      "z_index": 9,
      "source": "render_motion",
      "prompt": "Four slender woolly llama legs and lowered woolly head sniffing potato plant, padded toes not hooves, dark nails, partial body, night moonlight",
      "negative_prompt": "hooves, horse, full body, face centered",
      "alpha_strategy": "auto_segment",
      "blend_mode": "normal",
      "opacity": 1.0,
      "motion": "approach_then_sniff"
    }
  ],
  "plant_state": {
    "present": true,
    "stage": "post_flower_night"
  }
}
```

Timeline & Project Progress (May 19)

Phase 1: Screenplay (completed)

Phase 2: Explorative Storyboarding (completed)

Phase 3: Export of “Beat”-JSON, Evaluation of beats & key-value pairs (ongoing)

Phase 4: Export of “Workflow”-JSONs for Comfy UI, one for each beat (End of May)

Phase 5: Shot and Sound generation using “Workflow”-JSONs in ComfyUI (June)
-> FOCAL Course on Comfy UI early June

Phase 6: Compositing in After Effects / Premiere, export for exhibition (early July)

Exhibition

Presentation Concept:

- Film shown on old CRT TV / VCR as a loop.
- AI-generated images vs. “Analog” playback technology poses interesting contrast for an exhibition
- For the group presentation, sound will play over TV speakers
- Low Resolution might help with budgetary issues

